

Lesson 9

Homework 1 – Identify All Indexes

- Earlier, you learned about the `index()` method, which returns the position of the first occurrence of the specified value.
- Using what you've learned from this and previous lessons ...
- **Complete the following function that collects into another list called `result` ALL the positions where a value occurs.**

```
## Find all the indexes
```

```
# Any given list
```

```
my_list = ...
```

```
# Target value
```

```
value = ...
```

```
result = []
```

```
# ...
```

```
print(result)
```

Homework 2 – Sieve of Eratosthenes

In the third century B.C., the Greek astronomer Eratosthenes developed an algorithm for finding all the prime numbers starting at 2 up to some number N. For N = 10, the algorithm works by first writing down the following list:

2 3 4 5 6 7 8 9 10

Underline the first number in the list, then cross off every multiple of that number. For example, after performing this step of the algorithm, 2 would be underlined and every multiple of 2 would be crossed off:

2 3 ~~4~~ 5 ~~6~~ 7 ~~8~~ 9 ~~10~~

The process repeats: underline the first number in the list that is neither crossed nor underlined, then cross off its multiples. Eventually, every number in the list will either be underlined or crossed out, as shown below. The underlined numbers are prime.

2 3 ~~4~~ ~~5~~ ~~6~~ 7 ~~8~~ ~~9~~ ~~10~~

Homework 2 – Sieve of Eratosthenes

- **Apply this algorithm with lists to output all the prime numbers up to some user-provided integer upper bound N .**
 - *If you'd like a hint on how to start, check out the next page to see one way to start.*

Homework 2 – Sieve of Eratosthenes

''' Homework HINT! '''

```
N = int(input('N = '))  
sieve = [True] * (N+1)  
for i in range(2, N+1):  
    ...
```

Homework 3 – Potential Word

- **Write a program to identify if a user's input is a potential word.**
- A word is considered **potential** if:
 - It contains a **minimum** of 2 letters.
 - It contains only English letters (both uppercase, lowercase accepted).
 - It includes **at least** one **vowel**.
 - It includes **at least** one **consonant**.
- Output **Yes** if word is potential, otherwise, output **No**.
- **Notes:**
 - 'a', 'e', 'i', 'o', 'u', and their uppercases are **vowels**.
 - A **consonant** is an English letter that is not a vowel.
 - Check out the **isalpha()** method.
 - A **string** is similar to a list of letters/characters.
 - Problem inspired by <https://leetcode.com/problems/valid-word/>

Homework 3 – Potential Word

```
''' Homework HINT! '''
```

```
word = input('potential word?: ')
```

```
isPotential = True
```

```
...
```

```
print('Yes' if isPotential else 'No')
```

Locker Puzzle Coding Challenge

- Simulate the behavior of the 10 students toggling 10 lockers based on the rule shared in the brain teaser from today.
- Specifications:
 - You have **10 lockers**, all **closed** at the start.
 - Each locker can be either:
 - False (closed)
 - True (open)
 - For **each student** (from 1 to 10), toggle every locker number that is a multiple of that student's number.

Locker Puzzle Coding Challenge

- Expected output:
 - After all the 10 students have taken their turn, print:
 - Which lockers are **open** (by number, not index)
 - The final state of all lockers
- Extra:
 - Instead of 10 students, allow the user to specify any number of students.